

Lecture 6: Investment

Energy Economics of the Green Transition

Rasmus K. S. Berg

Monday 19 October 2026

Department of Economics, University of Copenhagen

Today

1. The model of ESM's section 7
2. The Model Lab and the Model Explorer
3. Part 1: read the reference system
4. Part 2: shock it, with carbon, fuel and capital
5. Part 3: margins of response

While we start: open the Model Lab and make your code,
championape-model-lab.share.connect.posit.cloud.



The model of ESM’s section 7

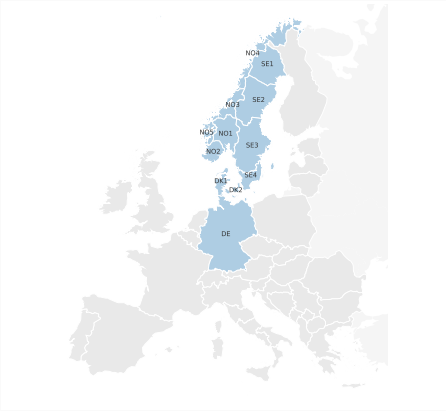
We now have all the ingredients

- **Dispatch** (lecture 3): the merit order; prices as marginal costs and scarcity rents.
- **Wind and solar** (lecture 3): variable output; capture prices and value factors.
- **Storage** (lecture 4): energy moved in time; the arbitrage band.
- **Flexible demand and trade** (lecture 5): demand that responds; zones joined by wires.
- **Investment** (today): capacity becomes a choice, at its annual cost.

Today, we are going to practise reading a full energy system analysis, and explain what happens and why.

The model: what should Denmark build?

- **The question:** what should Denmark build for a decarbonised power system in 2050, and how should it run it, at the lowest total cost?
- **Twelve zones:** DK1 and DK2, Germany, five in Norway and four in Sweden.
- **Denmark is built from scratch;** the neighbours keep today's plants and may add more.
- **One carbon price,** 85 €/t, everywhere.



Energy System Models, Fig. 6.1: the twelve zones of the model.

The Lab and the Explorer

The Model Lab

- **Like our worksheets:** today's questions, part by part; submit a part to see the keys and explanations.
- **With an AI tutor:** feedback on your open answers, which you can revise.
- **And an exercise bank** to study for the exam. If it works, it grows: earlier and later lectures, and exam-style questions with suggested solutions.

Your code is your login: write it down. On paper: the worksheet.



championape-model-lab.
share.connect.posit.cloud

The Model Explorer

- **Lectures 6 and 7:** how to analyse a (relatively) large energy system model.
- **The Explorer** collects its results: the reference case and about a hundred scenarios, each with one given changed.
- **Use it** to analyse how the energy system works, and how it responds when a given changes: compare a scenario with the reference.

championape.github.io/EnergyEconomics-explorer

Part 1

The new ingredient: the cost of capacity

$$F = \left(a(r, n) + \hat{f}\right) I$$

$$LCOE = \frac{F}{8760 \cdot CF} + c$$

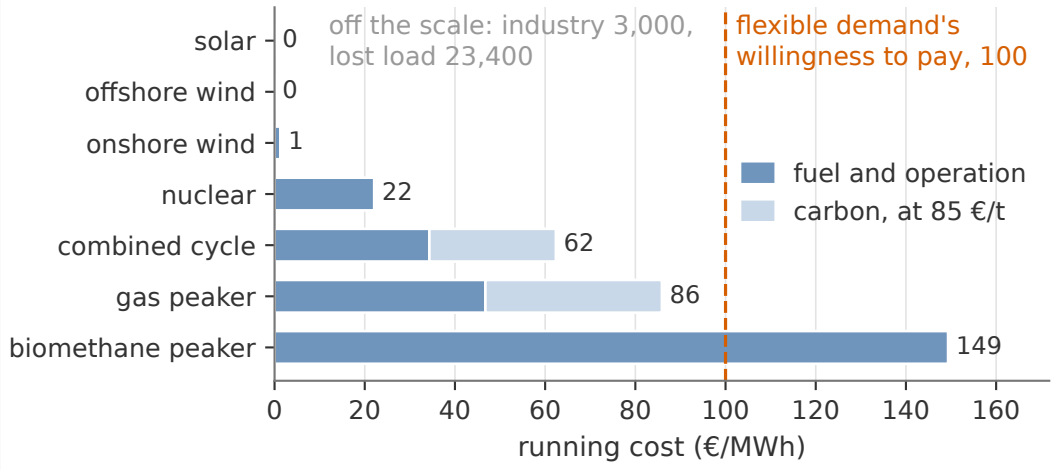
	F k€/MW a year	full-load hours	LCOE €/MWh
solar	29	953	30.8
onshore wind	94	3,060	32.0
offshore wind	168	4,508	37.3
gas peaker	45	938	95
combined cycle	100	–	–
nuclear	754	–	–

Source: ESM §7.1 and Table 5, at $r = 7\,\%$. The LCOE is before carbon, at the full-load hours, $8760 \cdot CF$, each technology runs in the reference system; –: not built.

The model: four kinds of participant

1. Generators
2. Demand
3. Storage
4. Interconnectors

The model: who sets the price



Source: Explorer (Money > Running costs, and demand's willingness to pay)

The model: welfare and the market

$$\max \quad W^{\text{inv}} \equiv \overbrace{\sum_h \sum_z \sum_j v_j \, d_{j,z,h}}^{\text{consumer value}} - \overbrace{\sum_h \sum_g \left(c_g + \tau e_g\right) q_{g,h}}^{\text{running costs, generators}} - \overbrace{\sum_h \sum_s c^s \left(g^c_{s,h} + g^d_{s,h}\right)}^{\text{running costs, storage}} \underbrace{- \sum_{g \in \mathcal{G}^+} F_g \, \bar{q}_g - \sum_{s \in \mathcal{S}^+} F_s \, \bar{s}_s}_{\text{investment and fixed costs}}$$

$$\sum_{g \in \mathcal{G}_z} q_{g,h} + \sum_{s \in \mathcal{S}_z} g^d_{s,h} + \sum_{\ell:\, b_\ell = z} \eta_\ell f_{\ell,h} = \sum_j d_{j,z,h} + \sum_{s \in \mathcal{S}_z} g^c_{s,h} + \sum_{\ell:\, a_\ell = z} f_{\ell,h} \qquad (\lambda_{z,h}) \quad \forall z, h$$

The model: buildable potential

$$0 \leq \bar{q}_g \leq \bar{Q}_g \qquad (\underline{\beta}_g, \overline{\beta}_g) \qquad \forall g \in \mathcal{G}^+$$

$$0 \leq \bar{s}_s \leq \bar{S}_s \qquad (\underline{\beta}_s, \overline{\beta}_s) \qquad \forall s \in \mathcal{S}^+$$

The potentials $\bar{Q}_g$ and $\bar{S}_s$ measure:

- the land and resources available;
- political constraints;
- any other limit that is not explicitly modelled.

The model: where a generator enters the Lagrangian

$$\mathcal{L}_g^{\text{inv}} = \underbrace{\sum_h \lambda_{z,h}\, q_{g,h}}_{\text{balance}} - \underbrace{\sum_h (c_g + \tau e_g)\, q_{g,h}}_{\text{running cost}} - \underbrace{F_g\, \bar{q}_g}_{\text{fixed cost}}$$

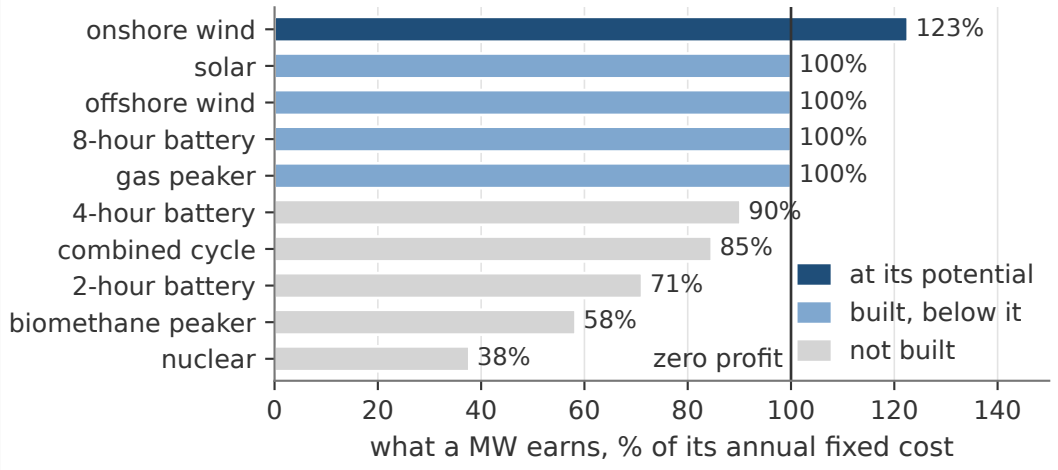
$$+ \underbrace{\sum_h \underline{\mu}_{g,h}\, q_{g,h} + \sum_h \overline{\mu}_{g,h}\big(\gamma_{g,h}\bar{q}_g - q_{g,h}\big)}_{\text{availability}} + \underbrace{\underline{\beta}_g\, \bar{q}_g + \overline{\beta}_g\big(\bar{Q}_g - \bar{q}_g\big)}_{\text{potential}}$$

Read the reference system

1. Open the lecture notes: *Energy System Models*, section 7.
2. Open the Model Explorer: *championape.github.io/EnergyEconomics-explorer*
3. Open the Model Lab: *championape-model-lab.share.connect.posit.cloud*

Solutions 1

Solutions 1, Q1: what a MW earns



Source: Explorer (Money > Rent per MW against annual fixed cost)

Solutions 1, Q1: Result 6, rents against fixed cost

$$\overline{\beta}_g - \underline{\beta}_g = \underbrace{\sum_h \gamma_{g,h} \max\left(\lambda_{z,h} - c_g - \tau e_g,\ 0\right)}_{\text{a MW's scarcity rent}} - \underbrace{F_g}_{\text{its annual fixed cost}}$$

Built, below its potential

$\overline{\beta}_g = \underline{\beta}_g = 0$

a MW earns exactly its fixed cost

At its potential

$\overline{\beta}_g > 0$

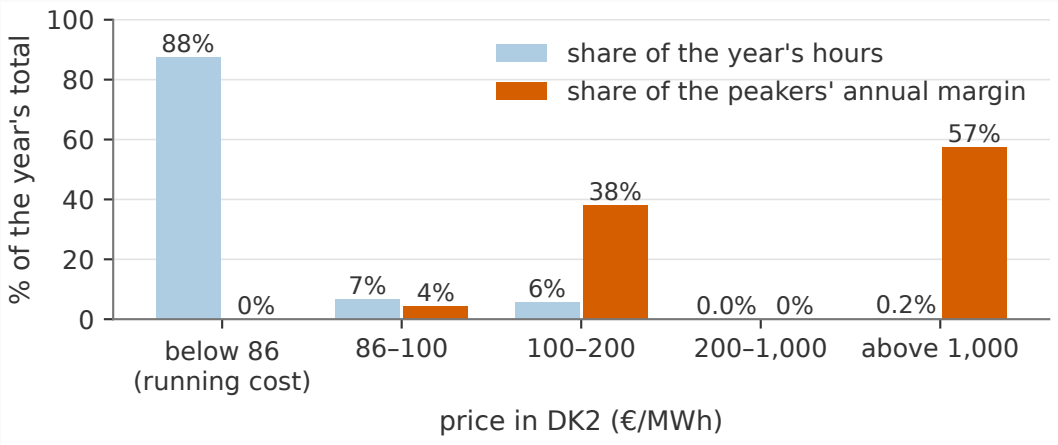
a land rent on the scarce sites

Not built

$\underline{\beta}_g > 0$

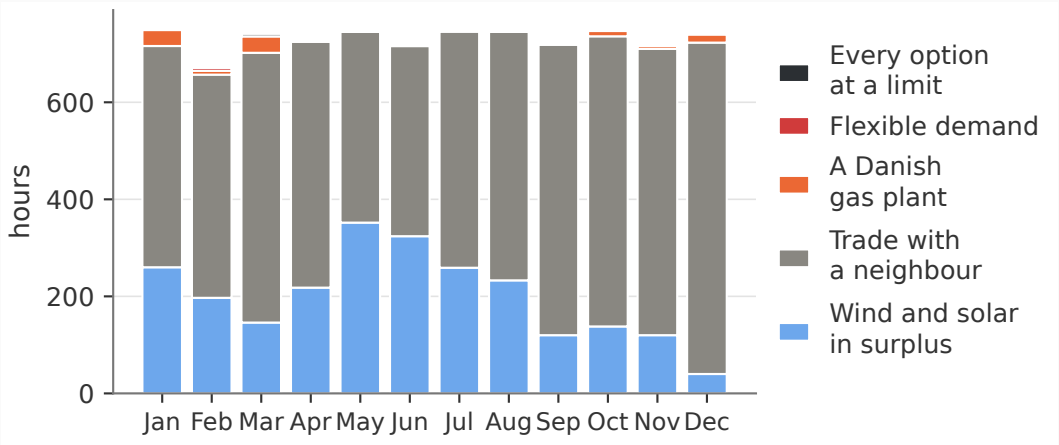
a MW would not pay for itself

Solutions 1, Q2: a peaker sells availability



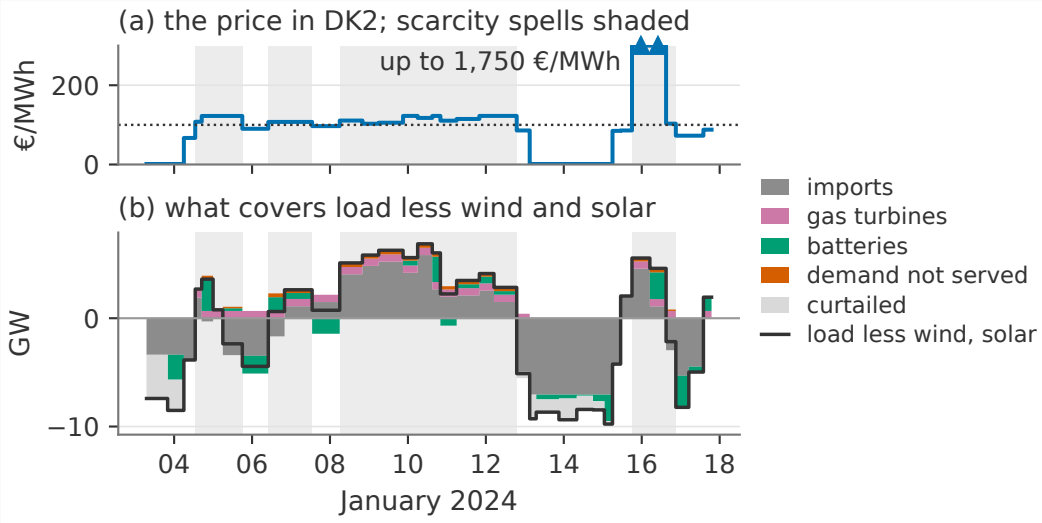
Source: Explorer (Money > Peakers' margin by price band)

Solutions 1, Q3: the Danish price is mostly made abroad



Source: Explorer (Operation > What is at the margin, month by month), DK2

Solutions 1, Q3: the highest prices ration what is left



Source: ESM Fig. 7.4 (a) and (b). Dotted: flexible demand's willingness to pay.

Part 2

Shock it: carbon, fuel and capital

1. We now analyse how the system changes under different scenarios: a different carbon price, more expensive natural gas, and cheaper capital.
2. The Explorer packages all relevant outcomes; your task is to sift through it to understand what happens and why.

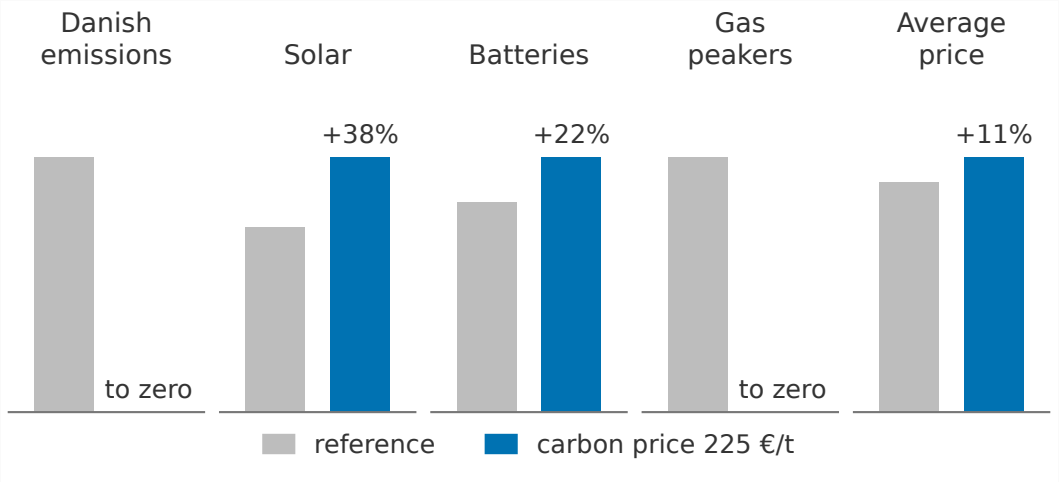
Solutions 2

Solutions 2, Q1–Q2: the questions

Q1. The carbon price rises from 85 to 225 €/t: Danish emissions, solar, batteries, gas peakers, the average price.

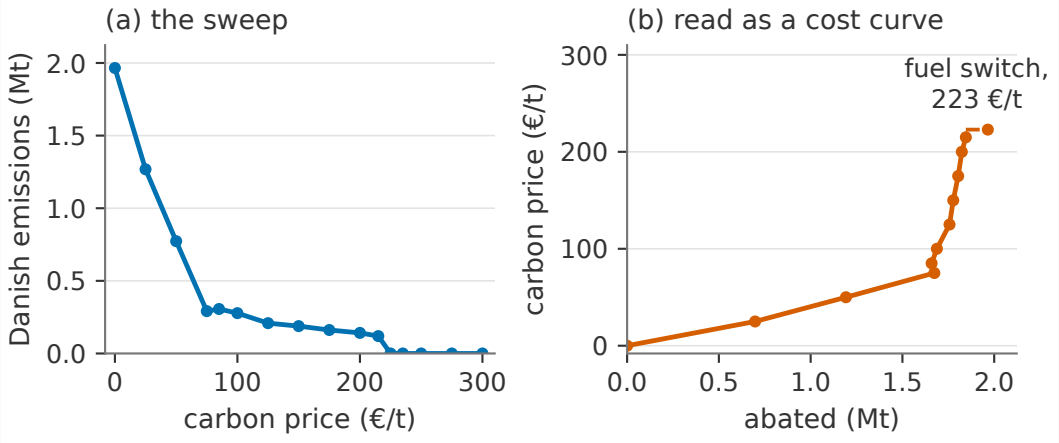
Q2. The shape of Denmark's marginal abatement cost curve.

Solutions 2, Q1: a higher carbon price



The Explorer's headline numbers, Carbon price 225 EUR/t against the reference; capacities in Denmark.

Solutions 2, Q2: Denmark's marginal abatement cost curve



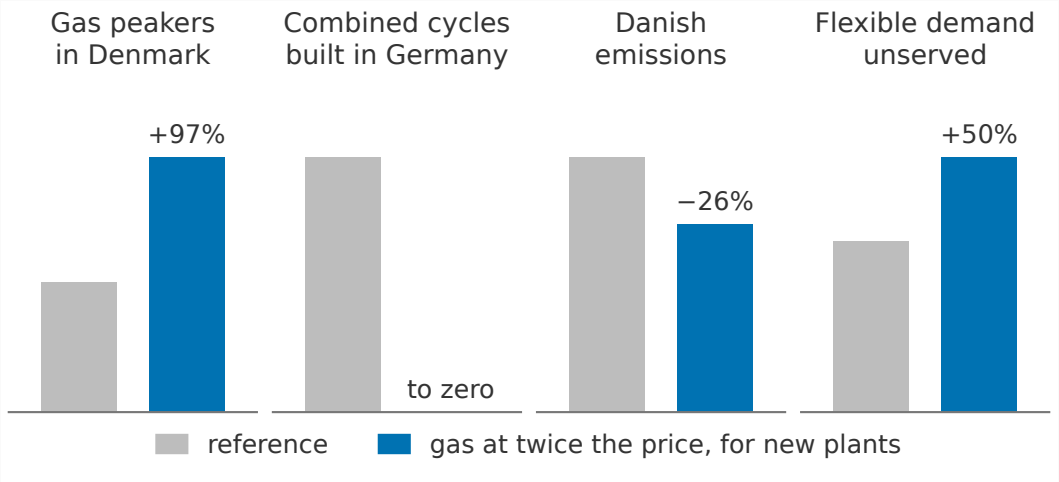
Drawn from the Explorer's data: (a) Sweeps, Danish emissions against Carbon price, every zone; (b) the same scenarios with the axes swapped.

Solutions 2, Q3–Q4: the questions

Q3. Gas doubles for new plants: predict; why Germany stops building combined cycles while Denmark builds more peakers; what takes gas's place.

Q4. Capital at 2 %: predict; why the discount rate matters more for wind and solar than for a gas plant.

Solutions 2, Q3: expensive gas



The Explorer's headline numbers, Gas at 2 x, new plants only, against the reference. ESM Fig. 7.6 shows it beside the other shocks.

Solutions 2, Q3: a gas plant has two jobs

Energy, over thousands of hours

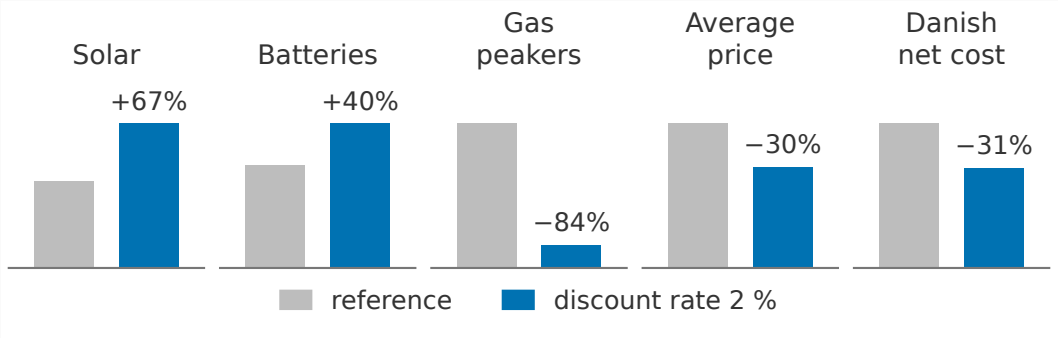
Once gas is expensive, wind, solar and batteries do it more cheaply: Germany builds no combined cycle.

Standing by, in winter spells

It came free with the combined cycles. Without them scarce hours come more often: Denmark doubles its peakers.

What takes gas's place: German solar and batteries for its energy; Danish demand response for its last hours.

Solutions 2, Q4: a system that is almost all capital



The Explorer's headline numbers, Discount rate 2 % against the reference; capacities in Denmark.

- **Wind, solar and batteries** cost almost only their annuity, which rises with *r*; a gas plant's cost is mostly fuel.
- **The cost of capital is the price level:** 95 % of the Danish bill is capital.

Part 3

Setup: margins of response

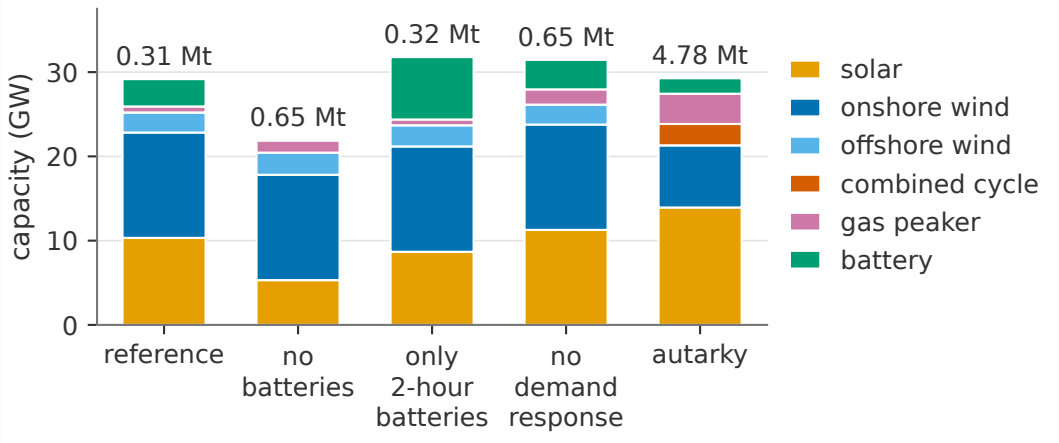
- **A system adapts to a shock through its margins of response:** building something else, storing, trading, or demand response.
- **Part 3 takes them away, one at a time,** to compare with the reference.

Margins of response

1. No batteries.
2. Only 2-hour batteries.
3. No demand response.
4. Autarky, and a Danish budget.

Solutions 3

Solutions 3: margins of response are substitutes



Drawn from the Explorer's data: Danish capacity, with Danish emissions above the bars. Take a margin away, and the others carry the adjustment, at a cost.

Wrap-up

What carries forward

1. **Prices recover capital.** What is built earns exactly its fixed cost; a scarce site earns a rent (Result 6).
2. **A renewable system's price is made by trade, surplus and scarcity.** The few expensive hours pay for the capacity that stands by.
3. **Margins of response substitute for one another,** at a cost; a territorial target leaks across a border.

Lecture 7, Monday 26 October: two drivers of the green transition, electric vehicles and hydrogen; and uncertainty. Read *Energy System Models*, sections 8 and 9.

Model answers to today's open questions: in the Lab tomorrow. The written solutions follow.

References i